BMC Medical Education

RESEARCH

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The effect of simulation training on the counseling skills of nutritional science students in dietetic internships

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Abstract

Simulations are a common tool in training aircrews, medical personnel and educators, providing experience in scenarios from the relevant content area and expanding the range of responses that simulation participants can acquire before they enter the field. Simulations have also been used in dietetic internships to improve clinical counseling skills. However, the use of structured simulations that involve progressive, multisession patient interactions has not been extensively studied in the context of dietetic internships. This study employed consecutive scenario models where interns developed ongoing interactions with the same simulated patient over multiple sessions, representing a significant methodological innovation in the field. A mixed method methodology was used to analyze closed and open-ended questionnaires which were administered to twenty-one interns, simulation actors and professional dietitians. This pilot study demonstrates the effectiveness of group and individual simulation training in enhancing dietetic interns' counseling skills during their training and describes how simulation can contribute to both the professional skills and self-efficacy required to lead a counseling session. These findings will enable additional academic departments to examine the integration of simulation into the training programs of interns in tracks that include a practicum.

Introduction

Nutrition therapy includes counseling patients for a wide range of complex health conditions. Like any counseling profession, it requires exceptional communication skills, empathy and problem-solving skills. These skills are developed through training that includes theoretical study, observations, and supervised practice programs (dietetic internships). One training methodology is the use of simulation, as proposed in this paper.

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Background

The use of simulations to develop skills is mentioned in professional literature [1, 2]. Simulation training, especially training that uses actors, has fostered experiential learning and accelerated skills development [3]. This approach aligns with Kolb's Experiential Learning Theory, which describes learning as a cyclical process involving four stages: concrete experience, reflective observation, abstract conceptualization, and active experimentation [4]. Simulations offer learners the opportunity to engage in real-time interactions (concrete experience), reflect on those experiences (reflective observation), derive insights (abstract conceptualization), and apply those insights in subsequent sessions (active experimentation). As such, simulation-based learning is well-suited to facilitate the development of complex interpersonal skills required in

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counseling and dietetics training. A clinical simulation session, that which includes an actor, is composed of three phases: briefing, simulation, and debriefing [5]. In the briefing phase, a trainee is given a scenario in which he or she will be trained. The trainee then participates in the simulation with an actor, and after simulation, the trainee is debriefed and receives feedback from the actor and by the group watching the simulation (in the case of a group simulation). In the study we describe in this article, the scenarios were meticulously prepared by a team of professional registered dietitians (RDs) according to established guidelines and delivered to the actors.

Simulations using actors have been found to be effective. A study by Ladousse [6] indicated that role-playing, an active learning technique, encourages counselors to think in real time and fosters adaptability and improvisation skills. Similarly, Nestel and Bearman [7] reported that using actors as simulated patients can help counselors improve their communication, problem-solving, and critical thinking skills.

In terms of counselors' self-confidence and perceived ability, Hargie et al. [8] reported that role-playing with actors can significantly increase self-efficacy and perceived communication ability, thereby improving the overall quality of counseling conversations. Similarly, another study has shown that practicing in a low-risk environment, counselors can experiment with different approaches, obtain immediate feedback, and hone their skills [9].

However, simulation with actors is not without challenges. The success of this approach largely depends on the ability of the actors to convincingly replicate patient behaviors [7]. Moreover, managing the logistical and financial aspects of organizing such simulation sessions may pose challenges for some institutions [10].

The benefits of simulation training in health care professions have been recognized in different areas. In medical education, simulations help learners develop technical and interpersonal skills in a safe environment [11, 12]. These studies provide a basis for examining the potential use of simulations in training counselors for consultations. Various forms of simulations, including virtual reality (VR) and actor-based simulations, have been utilized to enhance learning outcomes [13, 14]. Simulations offer immersive experiences, allowing students to engage in realistic scenarios that mimic clinical settings and improve clinical reasoning, decision-making, and interdisciplinary communication skills [10]. Among dietetics students, actor-based simulations have effectively developed communication, problem-solving, and critical thinking skills [15].

Simulation-based training has emerged as a valuable tool in dietetics education, bridging theoretical knowledge and real-world clinical practice. The particular use of simulation to train nutritionists in counseling sessions can reproduce a wide range of scenarios with patients, preparing trainees for the variety of cases they may encounter in their work as nutritionists [16]. In addition, feedback from simulations can be immediate and adapted to trainee performance, enabling focused learning and improving skills [10].

Buchholz et al. [17] investigated the impact of simulated patients (SPs) on dietetics students' communication and nutrition-care competence at the University of Guelph. Their study involved 17 undergraduate and 15 graduate students using SPs portrayed by undergraduate theatre students. These simulations covered various health conditions with increasing complexity, and assessments based on Canadian Integrated Competencies revealed significant improvements: undergraduate students improved by 49.7% in communication and 45.8% in nutrition-care skills, whereas graduate students improved by 18.5% and 37.9%, respectively. This study concluded that actor-based simulations effectively enhance practical competencies in dietetics education.

Similarly, O'Shea et al. [18] conducted a systematic review of simulation-based learning experiences (SBLE) in dietetics programs. They reviewed primary methods such as the use of SPs to develop and assess communication and counseling skills through realistic patient interactions. Their review highlighted the use of objective structured clinical examinations for competency-based assessments, where students rotated through stations with specific tasks or clinical scenarios involving SPs. Interprofessional education, which fosters collaboration among students from various health care disciplines to enhance teamwork and communication skills, was also emphasized. The authors concluded that while SBLE are valuable, there is a need for more quality and quantity SBLE research to ensure that these simulations are pedagogically sound and have a measurable impact on learning outcomes [18].

Notably, existing research has not systematically investigated consecutive scenario models where interns develop ongoing interactions with the same simulated patient over multiple sessions, representing a significant methodological gap in the field.

Our study aims to fill the identified gap by providing a structured and comprehensive approach to integrate simulation into dietitian trainees' internships. By incorporating realistic and emotionally charged scenarios within a well-organized framework, our study seeks to better prepare dietitian trainees for the complexities of real-world clinical interactions.

Internship structure for dietitians in Israel is regulated by the Israeli Ministry of Health, which requires students to complete 84 h of supervised practical training in community clinics. During these supervised sessions,

the intern conducts an intake of dietary history, provides nutritional guidance, and drafts meal plans. These sessions are supervised by experienced registered dietitians, who serve as internship tutors, professionals responsible for mentoring, guiding, and evaluating interns during their practical training. Despite this active involvement, most training hours are dedicated only to observation. Even when the intern is actively involved, the interaction is supervised by a registered dietitian, which limits opportunities for one-on-one intimate patient interaction. Given the time constraints often present in clinical settings, adding simulation-based training to the internship process, alongside hands-on field experience, enables interns to engage more extensively in active learning within a safe and controlled environment, while receiving structured feedback.

As part of a pilot program to evaluate the integration of simulation into dietetic internships, we received regulatory approval to substitute simulation training for 30% of traditional field experience hours.

Therefore, it is important to examine whether the incorporation of well-designed simulations has added value as part of the internship before and during field experience.

Additionally, exploratory data on different types of training, including complex and continuous scenarios, along with feedback from the actors and the mentor during training, remain limited. Our study aims to address these gaps by examining the perceived effectiveness of clinic-like simulations conducted by actors and designed to mimic real patient interactions in a developing scenario over three consecutive sessions with the same patient. An added advantage of actor-based simulations is their ability to emotionally engage participants, allowing their professionalism to be tested under emotional stress, which can be critical in real-world clinical settings. By focusing on the immediate impacts of these simulations, our research provides valuable insights into their effectiveness in enhancing professional competence and practical skills.

The current pilot program examined a model that divides training between "traditional" clinical observations and simulations with actors, followed by analysis and feedback from peers and professionals. The focus of this study was to identify the effectiveness of simulation use in dietetic internship programs and its impact on dietitians' counseling skills in future counseling sessions.

From this goal, the following research questions were derived:

1. How do interns perceive the effectiveness of the training through group simulation?
2. How does personal simulation affect the interns' ability to conduct a counseling session, as perceived by the interns, actors, and internship mentors?
3. How do group and personal simulations affect interns' self-efficacy?

Methodology

Research design and population

The study population consisted of nutritional science students in their final year of academic studies in the Department of Nutrition at Tel Hai Academic College (interns). The simulation training consisted of two components: [1] Group simulations: Three group workshops, each containing three simulation rounds (total: 9 group experiences) [2], Personal simulations: two individual one-on-one sessions.

Simulation sessions methodology

In the group simulations, the interns attended 3 four-hour workshops, in which they learned about communication skills and then experienced scenarios that simulated situations of clinical conditions and patient types they might encounter in their clinical work. The workshop scenarios were meticulously designed to simulate real-life clinical situations, with a focus on three specific patient cases: Saxenda (Liraglutide) treatment, dyslipidemia, and bariatric surgery. Each scenario was structured as a continuous storyline, beginning with an initial patient consultation (intake) followed by two subsequent sessions to assess and address compliance and outcomes. These sessions were specifically tailored to challenge young dietitian interns by introducing them to patients with diverse personalities and complex social backgrounds, often as old as their parents or grandparents. This approach exposes the interns to the emotional regulation and life challenges commonly encountered in clinical practice, which sometimes prove more challenging than the clinical cases themselves. Interns were required to address patient objections and a general lack of basic knowledge in nutrition and health, thereby enhancing their ability to navigate emotionally charged interactions effectively.

The structure of the group simulations is described in Table 1. Interns participated in 3 workshops, each with 3 rounds, each round composed of three phases: briefing, counselling session simulation, and debriefing. A model of a round is described in Fig. 1 (stages 2–6).

Group simulation workflow showing the briefing-simulation-debriefing cycle for individual rounds within workshops is described in Fig. 1.

Workshop 1: obese patient using GLP-1 analog

Round 1: Initial dietitian consultation: preparation for treatment; Round 2: 1-month follow-up, addressing

Table 1 Detailed structure of group simulation workshops

Workshop	Patient Case/Focus	Rounds of counseling sessions (Individual Interactions)
1	An obese patient usingGLP-1 analog	Round 1: Initial dietitian consultation; Round 2: 1-month follow-up; Round 3: 3-week follow-up
2	Patient with high blood lipid (TG, LDL) levels	Round 1: Initial consultation; Round 2: 3-month follow-up (patient missed 1- & 2-month); Round 3: 2-month follow-up (patient missed another)
3	Bariatric patient (4 years after surgery, weight regain)	Round 1: Post-bariatric surgery evaluation; Round 2: 1-month follow-up; Round 3: 3-week follow-up

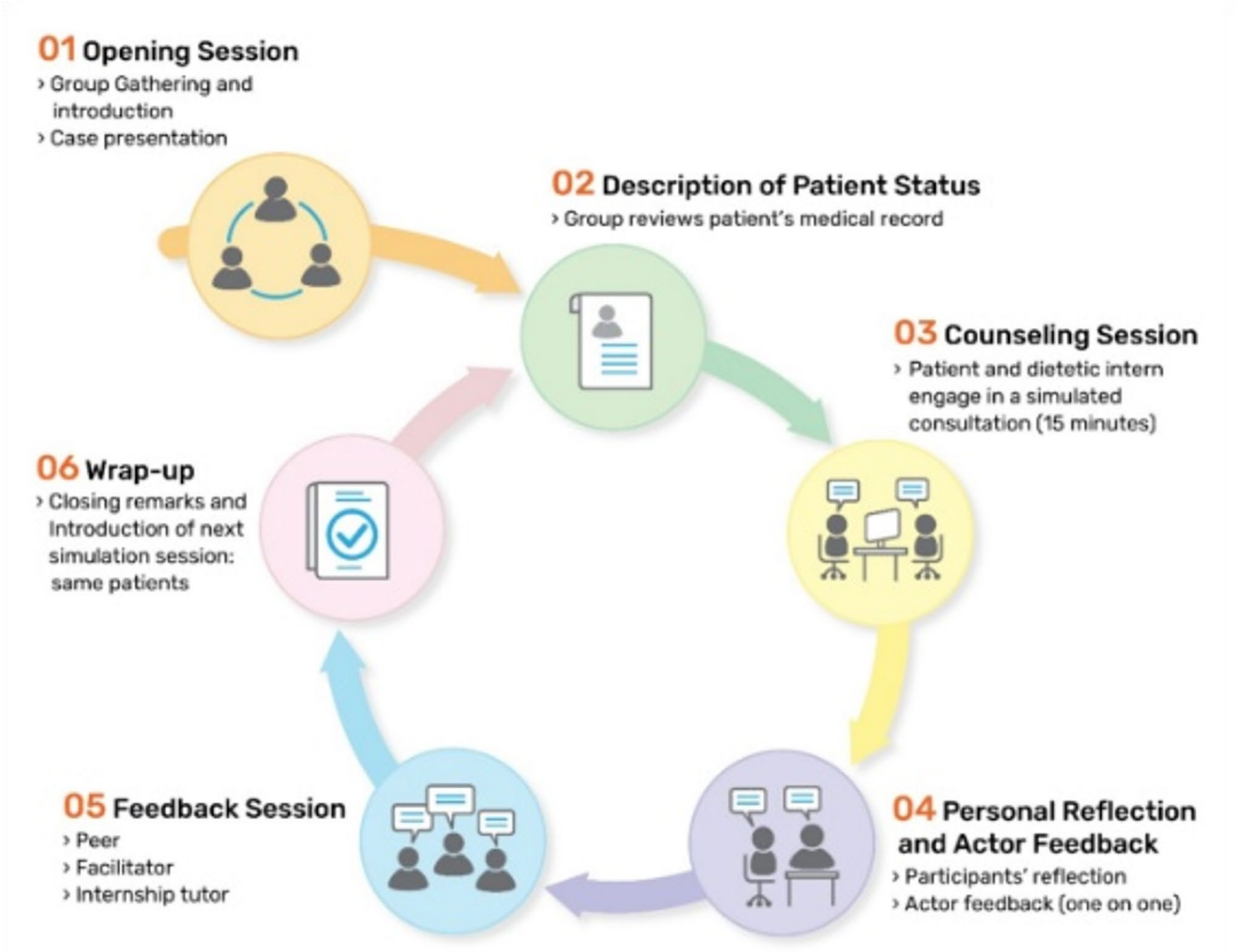


Fig. 1 Group simulation round workflow

challenges and adjusting treatment; Round 3: 3-week follow-up, managing over adherence and unrealistic expectations.

Workshop 2: patient with high blood lipid (TG, LDL) levels [evaluated at T1

Round 1: Initial consultation: overcoming resistance and structuring dietary habits; Round 2: 3-month follow-up, patient missed 1- and 2-month follow-up. Addressing misconceptions and reinforcing positive changes; Round 3: At the 2-month follow-up, the patient missed another follow-up visit. Balancing dietary changes and addressing social and digestive challenges.

Workshop 3: bariatric patient (4 years after surgery, weight regain) [evaluated at T2

Round 1: Post-bariatric surgery evaluation and addressing nutritional deficiencies; Round 2: 1-month follow-up. Addressing unrealistic weight expectations and reassessing dietary adjustments; Round 3: 3-week follow-up. Focusing on overall health beyond weight loss.

A detailed description of each case scenario is provided in Appendix A.

Group simulations were led by a simulation mediator with expertise in conducting simulations. In each simulation workshop, three trainees participated in the actual simulation with an actor-patient. Each simulated counselling session was followed by structured debriefing sessions lasting 20–30 min, incorporating: [1] self-reflection by the intern [2], multi-source feedback phase including structured input from the simulated patient (actor), observing peers, supervising registered dietitian, and simulation facilitator [3], analytical discussion examining clinical reasoning and communication strategies, and [4] application phase connecting insights to future clinical practice. This 360-degree feedback approach ensured comprehensive learning from multiple perspectives. Surveys were conducted between sessions and at the end of the simulation session. The process was further reinforced with follow-up meetings by a professional dietitian to ensure that learning was consolidated.

Simulation environment and infrastructure Simulations were conducted in a dedicated room designed to resemble a typical outpatient clinic setting. The simulation room was equipped with audio-visual recording technology, including multiple cameras and microphones positioned to capture all interactions. Participating interns observed real-time simulated counselling session from an adjacent observation room through live video feed, allowing for immediate peer learning and preparation for subsequent debriefing sessions.

Facilitator Qualifications and Actor Preparation: All simulation sessions were led by a trained simulation facilitator, and professional actors portraying simulated patients underwent a full-day standardized training conducted jointly by the supervising registered dietitian and professional simulation facilitator. The training protocol included: [1] detailed character background and medical history briefing [2], clinical scenario familiarization and role-play practice [3], standardized response protocols and emotional consistency training, and [4] feedback delivery techniques for post-simulation debriefing. Actors strictly adhered to predetermined scripts and character profiles to ensure standardization across all simulation encounters.

In addition to group simulations, the interns participated in two *personal simulations* of counseling sessions, each lasting 20 min, approximately two months after the group simulation workshops. In these sessions, the actors portrayed patients with gastroenterological issues, diabetes, and metabolic syndrome, conditions selected owing to their high prevalence in clinical practice. This focus on common conditions aligns with the goal of providing interns with practical experience in managing frequently encountered patient cases [19]. Each simulation was recorded, and following the sessions, the interns received immediate feedback from the actors. These recordings were subsequently reviewed by a senior clinical dietitian, who provided detailed, one-on-one feedback to reinforce the interns' learning experience.

The personal simulation model including the individual counseling session structure with immediate actor feedback and subsequent tutor review is described in Fig. 2.

Scenario development The development of simulation scenarios incorporated established best practice prin-



Fig. 2 Personal Simulation workflow

ciples for simulation design, including alignment with learning objectives, scenario fidelity, and structured debriefing. Each scenario was co-developed by faculty and experienced clinical dietitians to reflect realistic clinical encounters, with an emphasis on common conditions encountered in outpatient care. The scenarios were constructed as progressive, multi-session cases with recurring simulated patients, enabling learners to experience continuity of care, emotional complexity, and communication challenges. A detailed description of each case scenario, including character background and learning objectives, is provided in Appendix A.

Research tools

The research was conducted using self-report questionnaires [20]. The multiple-choice questions were analyzed using quantitative methodologies of frequencies and averages, and the open-ended questions were analyzed with qualitative methodology, where the answers were categorized by themes.

Questionnaires

All questionnaires were administered via Google Forms. In the group simulations, the questions were multiple choice, and participants could choose more than one answer.

The interns' pre- and post-personal simulation questions included multiple-choice questions for which we used a 7 item Likert scale (7-very much agree, 6-greatly agree, 5-slightly agree, 4-moderately agree, 3-moderately disagree, 2-slightly disagree, and 1-strongly disagree). In addition, the questionnaires for the actors and internship tutors also included open-ended questions for which we employed a descriptive approach. This exploratory nature of the research can provide valuable insights into tutors' and actors' attitudes and perceptions.

Assessment of group simulation

The interns' learning experience from the group simulations was assessed before and after the simulation using 15 statements divided into three topics: [1] preparation for the simulation workshop [2], relevance and learning during the workshop, and [3] insights following the workshop. Since the statements in the questions in this questionnaire were not ordinal, the questions were analyzed by frequency of intern responses regarding each part of the simulation workshop.

Assessment of personal simulations

Personal simulations were evaluated in three ways: [1] **Interns' self-perceived ability** to engage with simulated patients was evaluated before and after each simulation. The questions in the pre- and post-questionnaires examined self-efficacy and counseling skills. The statements included in the questionnaires were adapted from tools developed by Barma and Friedman [18], which focus on perceptions of professional development and reflective practice in educational settings. These were selected for their relevance to the internship context and their demonstrated validity in similar learning environments. In addition, the questionnaire design aligns with Bandura's [20] theory of self-efficacy, emphasizing the role of personal belief in one's ability to perform tasks and achieve goals. This theoretical foundation provided a strong rationale for assessing interns' perceptions of competence, motivation, and learning processes during simulation-based training and field experiences. The questionnaire statements were adapted to the simulation and validated by a focus group [2]. **Actor feedback** on the students' professional performance and communication skills. The actor questionnaire included 5 multiple-choice questions, for which we used a 5 item Likert scale (5-very much agree, 4- agree, 3-moderately agree, 2-slightly disagree, and 1-strongly disagree), and [3] **Internship tutors' perceptions** of the intern's professional performance and communication skills. The tutors were two experienced RDs (with more than 15 years of seniority) accredited for training interns, who viewed the simulation recordings and evaluated the interns' performance using a questionnaire of 6 multiple-choice questions using a 5 item Likert scale – (5-very much agree, 4-agree, 3-moderately agree, 2-slightly disagree, 1-strongly disagree).

Data analysis

Group simulation

Twenty-one interns in the Department of Nutrition experienced groups of simulations.

Personal simulation

Twenty interns took part in personal simulations. To test the effect of the simulation, we compared the differences before and after the simulation. Since the actors' and tutors' closed-ended questions in the questionnaires were on a scale of 1–5, and we wanted to compare the evaluation to that of the students, the answers of the student questionnaires in the personal simulation (pre- and post-) were re-coded to a scale of 1–5 (Table 2).

Self-efficacy before and after the personal simulations were examined using paired-samples t-tests. The interns were identified by the last four digits of their mobile phones.

Actor's feedback

The actor's evaluation of the interns' skills and professional performance was assessed by a post-simulation questionnaire filled by the actor. The questionnaire included open and closed questions on a five-point Likert like scale (with verbal assessment as presented in

Table 2 Conversion table for the agreement scales in the closed questionnaires

Measures on a scale of 1–7	Verbal assessment on a scale of 1–7	Converted to a scale of 1–5	With verbal assessment on a scale of 1–5
7	Agree	5	Strongly agree
6	Somewhat agree	4	Agree
5	Either agree or disagree agree	3	Neither agree nor disagree
4	Somewhat disagree	3	Neither agree nor disagree
3	disagree	3	Neither agree nor disagree
2	Disagree	2	Disagree
1	Strongly disagree	1	Strongly disagree

Table 3 Percentage of statements related to aspects of learning from the group simulation

	Statements	Percentage of all statements
Statements relating to aspects of learning	I learned a new theory	78.0%
	New avenues of thought opened to me	73.2%
	I learned new skills	53.7%
	My thinking and the connections between things became more organized	26.8%
Statements relating to aspects of applying learning	I will know how to use, infer and apply different tools according to the various incidents I will encounter	56.1%
	I expanded my repertoire of abilities. I believe I will know how to respond in a similar instance	24.4%
	I was presented with new tools and ways of thinking	22.0%

Table 2). For the open ended questions, we used a thematic analysis approach to identify patterns of meaning.

Tutors' perceptions of the personal simulation

After the personal simulations, internship tutors, senior clinical dietitians who are faculty members from the Department of Nutrition at the College, reviewed 26 simulation recordings. Nine interns were viewed in two simulations each, whereas eight students were viewed in only one simulation. After watching each recording, the tutors completed a questionnaire assessing aspects of the interns' consultation procedures and the professional knowledge they demonstrated.

This questionnaireincluded closed questions on a five-point Likert like scale (with verbal assessment as presented in Table 2), and two open-ended questions. The open ended questions, were analyzed using thematic analysis approach to identify patterns of meaning.

Ethics

The study was conducted following the principles outlined in the Declaration of Helsinki. All participants were fully informed of the objectives and procedures of the research, their rights to anonymity and confidentiality, and the voluntary nature of their participation. Written informed consent was obtained from all participants. The research protocol was reviewed and approved by the Ethics Committee of Tel Hai Academic College (approval number: June 38 – 6/2023). Clinical trial number: not applicable.

A completed STROBE-SIM checklist [21], mapping the manuscript content to the recommended reporting items for simulation-based research, is provided in Appendix B.

Findings

Group simulations

Twenty-one interns experienced group simulations (the total number of responses to the group simulation questionnaires was 41). A total of 98% of the respondents described the experience as beneficial. The relevance of the simulation workshop to the interns' professional work was beneficial: 98% of the responses indicated that the participants found the simulation workshop scenarios to be relevant to them. The interns were asked to mark statements related to learning aspects from the simulation. Since it was possible to mark more than one statement related to these aspects, the table shows the percentage of interns who chose each statement. 78% of the students felt that they had learned a new theory and were exposed to new directions of thought. In terms of applying learning, more than 56% of the interns felt that they would know how to use, infer and apply different tools acquired in the simulation in the future (Table 3).

Personal simulations

An examination of the interns' experience with the personal simulation revealed that the simulation scenarios were very relevant (AVG = 4.65, SD = 0.54) and that the interns benefited from the simulation consultation (AVG = 4.49, SD = 0.69). In addition, two skills were developed during the personal simulations:

Professional skills

The findings show that personal simulation contributed to the development of professional skills to a great extent (Table 4), as perceived by the interns, the actor and the tutors:

Table 4 Contribution of personal simulation to interns' professional skills

	Statements referring to professional skills	M	SD
Interns' self-evaluation (*N* = 37)	After the simulation I expanded the repertoire of my responses	4.51	0.65
	The simulation allowed me to express my professional knowledge	4.27	0.73
	The simulation contributed to my self-confidence for my future consultation sessions	4.43	0.73
Actors' ("patient") evaluation (*N* = 43)*	Confidence in the response I was given in the consultation	4.40	0.82
	A clear explanation.	4.23	0.87
	I felt tension, anxiety, or a lack of confidence in the student	1.88	1.12
Tutors' evaluation (*N* = 26)	The intern displayed a high level of professional knowledge	3.54	1.24
	The Intern related comprehensively to the patient	4.27	1.08
	The intern answered questions that arose during the session	4.27	0.96
	The intern explained clearly and in appropriate language	3.96	1.08

*During the first five personal simulations, the students did not complete pre-and post-feedback, but the actors did, and so there are five additional responses for the actors

Table 5 Contribution of personal simulation to the interns' communication skills

	Statements referring to professional skills	M	SD
Interns' self-evaluation (*N* = 37)	The simulation enhanced my listening skills	4.46	0.65
	The simulation enhanced my ability to express empathy	4.54	0.61
	The simulation enabled me to adapt my messages to the character of the patient	4.43	0.65
	The simulation improved my ability to explain and express myself to the patient	4.51	0.65
Actors' ("patient") evaluation (*N* = 43)*	The students' interpersonal skills advanced the consultation conversation	4.26	0.95
	A sense of empathy for the student	4.30	0.91
Tutors' evaluation (*N* = 26)	Organizational ability and leading the session	3.46	0.99
	Relating to the patient during the session	5.00	0.00
	Intern's ability to cope with resistance during the consultation	4.65	0.85

*During the first five personal simulations, the students did not complete pre-and post-feedback, but the actors did, so there are five additional responses for the actors

Professional skills such as knowledge and the ability to answer questions clearly and in appropriate language during the session (as reported by the interns, actors, and tutors) were developed. Confidence in the consultation was also developed (as reported by the interns and actors).

In the open-ended questions, the tutors referred to the interns' ability to take patient anamnesis:

"[The intern] is very pleasant and made efforts to connect with the patient and ask the relevant questions."
"Many questions were asked, on the basis of which the case can be well understood."

In three cases of viewing the same intern twice, the tutors noted an improvement in the ability of the intern to conduct the session:

"Improvement in anamnesis and patient attitude."
"Very significant improvement in the ability to create a connection."
"Significant improvement in attitude toward the patient, inclusive."

Communication skills

The findings from the questionnaires show that personal simulation largely contributed to the development of communication skills (Table 5), as perceived by the interns, the actor and the tutors:

In the open-ended questions, the tutors described examples of the simulation's contribution to communication skills:

"The intern was very pleasant throughout the conversation and seemed to be able to enlist the patient in further treatment by complying with the nutritional recommendations despite the patient's reservations. She seemed to speak to her in an inviting way so they could continue to make meaningful connections going forward."
"His attitude is very respectful, inclusive and engaging."
"[She was] pleasant and inclusive despite the difficulty in engaging the patient."

Simulations' contribution to self-efficacy

An examination of the self-efficacy parameters before and after the personal simulation revealed significant

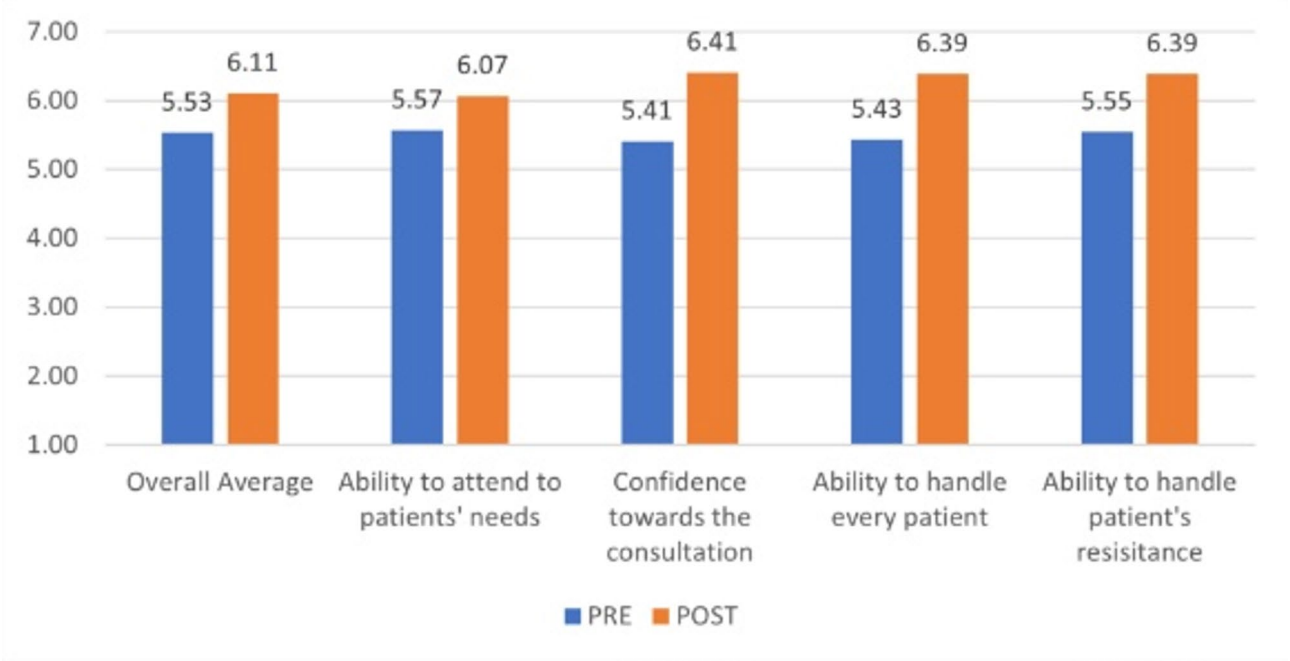


Fig. 3 Significant differences of interns' feelings in personal simulations (all $p < 0.01$)

differences in the interns' sense of confidence in their professional knowledge in their belief that they know how to respond to patients, in their confidence that they felt in preparation for consultations, and in their belief in their ability to cope with each patient and with patient resistance during the consultation (Fig. 3).

Discussion

This pilot study used structured groups and personal simulations of consecutive scenarios of the same patient and aimed to examine the effectiveness of the simulations on the interns' ability to conduct a counseling session and the effect on the interns' self-efficacy. We found that self-efficacy in terms of professional skills and communication skills was promoted via simulation training, as reported by the interns themselves.

Simulation training provides interns with a safe and controlled environment that allows participants to develop and practice their skills [22, 23], thus preparing them for the variety of cases they may encounter in their practice [3]. The effectiveness of simulations is derived, among other things, from the choice of scenarios [24]. In this study, the interns felt that the topics of the selected scenarios were largely relevant to their fieldwork in clinical settings and that the group simulations improved their learning and their ability to use, learn from, and apply tools they received during the course (Table 2).

The use of consecutive simulations allowed a more authentic replication of real-life patient interactions. It allowed participants to accumulate knowledge gained from previous sessions and use it in the subsequent meeting, thereby enhancing their ability to manage difficult clinical scenarios that require emotional regulation and professional composure. For example, scenarios included dealing with adherence challenges, late arrivals for follow-up sessions, lack of family support for treatment goals, and cross-cultural communication challenges - situationswhich may not typically be covered in theoretical dietetic education, where the focus is more on nutritional recommendations than practical challenges of patient interactions. Facing these challenges and acquiring skills essential for dietitian's work is crucial for delivering effective nutrition counseling. Importantly, the settings and responses used in these simulations were derived from real patient cases encountered by the team of dietitians who developed the scenarios. This ensured that the simulations were realistic and directly relevant to the interns' future practice. In addition, feedback from simulations is immediate and relevant to the intern's performance, enabling focused learning and skills improvement [9].

The simulations enabled the interns to improve their sense of professional skills, which is expressed in their sense of confidence in their ability to provide nutritional therapy, in expanding their repertoire of responses, and in their ability to explain dietary guidelines. In terms of conducting counseling sessions, the interns also reported that they felt an improvement in their ability to lead the meetings calmly.

Simulation training can improve communication skills [2]. In this study, the interns reported that the simulation improved their ability to express empathy and to

communicate with patients. The improvement in the ability to address resistance was also noted by the internship tutors, who viewed the simulations. However, no significant differences were found in the pre- and post-simulation questionnaires or in the interns' confidence in their ability to better prepare for the consultations.

The methodological framework developed in this study addresses documented implementation challenges in simulation-based dietetic education. Research has identified significant barriers to multi-session simulation implementation, including resource constraints, curriculum integration difficulties, and limited faculty development [25]. Our consecutive scenario approach demonstrates successful integration within existing educational structures.

We showed that participation in two group simulation followed by two personal simulation as a training tool is effective in the promotion of professional self-efficacy and communication skills, but it is not without limitations.

The pilot was built to provide a multidisciplinary and integrated training solution that included a pedagogical team that led the simulation process along with a professional academic team that offered a professional perspective as well as a connection to the clinical field.

While our study does not address long-term impacts, it significantly contributes to understanding the short-term benefits and practical application of simulation-based training in dietetics education, demonstrating its potential to enhance immediate professional readiness. In addition, this study has several limitations. First, this is a pilot study (or proof of concept) with a relatively small sample of 20 students. Second, the students who volunteered to participate in the pilot may have been more adept in the material and exhibited higher levels of self-efficacy, potentially introducing selection bias. The limited sample size may restrict the generalizability of the findings. Third, this study focused on short-term impacts immediately following the simulation training and did not assess long-term retention of skills or transfer to actual clinical practice. Additionally, the absence of a control group limits comparative analysis. Despite these limitations, this pilot study provides valuable preliminary evidence for the potential effectiveness of structured simulation training in dietetic education. The results can inform the design of future larger-scale studies to further validate the effectiveness of simulation-based training in dietetic internships.

The strengths of the study include a high response rate - out of 23 interns who participated in the pilot, 20 interns answered the research questionnaires. In addition, information regarding the effectiveness of the simulation was obtained from three different evaluations: a subjective assessment of the participating interns, an evaluation of the internship tutor and an evaluation of the actor. A combination of the different perspectives provides comprehensive information about the effectiveness of the tool.

In addition, using a continuous scenario that imitated real-life patient relationships enhanced the simulation process by providing a realistic and immersive training environment for the interns.

Summary

This study demonstrates the effectiveness of using consecutive group simulations followed by personal simulations as part of the dietetic internship, enabling interns to experience the application of the knowledge acquired during their studies in a controlled and protected environment prior to entering the field.

The unique approach to preservice training of dietitians, which uses group and personal simulations with progressive scenarios and consecutive sessions, has allowed us to bridge the gap between theoretical knowledge and practical application. The simulations, which were intended to depict real-life challenges and were followed by immediate personalized feedback, helped develop both professional and interpersonal skills. These findings provide valuable preliminary evidence for the potential of consecutive simulation training in dietetics education and suggest that future research could further validate this approach.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12909-025-07674-x.

Supplementary Material 1: Appendix A. Case scenario descriptions.

Supplementary Material 2: Appendix B. STROBE-Sim Compliance Checklist.

Author contributions

Ifat Linder, Zafit Lukach, Sigal Tepper and Keren Pergament designed the study. Keren Pergament, Rafi Steckler, and Sigal Tepper designed the scenarios for the simulations. Zafit Lukach analyzed the data, and all authors contributed to writing the manuscript, approved it, and agreed to be accountable for all aspects of the manuscript.

Data availability

The datasets generated and/or analyzed during the current study are available from the corresponding author upon request.

Declarations

Competing interests

The authors declare no competing interests.

Received: 14 April 2025 / Accepted: 9 July 2025
Published online: 25 July 2025

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